

Department of Mathematics and Applied Mathematics

Module 3MS: Metric Spaces 2023

Class test 1 solutions

Total Marks: 40

1. (a) i. $\overline{A} = \{(a, b) \in \mathbb{R}^2 : 0 \leq a \leq 2095\}$
ii. $\text{bd}(A) = \{(a, b) \in \mathbb{R}^2 : a = 0\} \cup \{(a, b) \in \mathbb{R}^2 : a = 2095\}$
iii. $\text{int}(A) = \{(a, b) \in \mathbb{R}^2 : 0 < a < 2095\}$
(b) i. $\overline{B} = B = \{(a, b) \in \mathbb{R}^2 : -10 < a < 1\}$
ii. $\text{bd}(B) = \emptyset$
iii. $\text{int}(B) = B = \{(a, b) \in \mathbb{R}^2 : -10 < a < 1\}$ [3]

Mark allocation for the above question:

Number of correct answers	Mark allocated
0	0
1 – 3	1
4 – 5	2
6	3

2. (a) For $x, y, z \in X = \{x \in \mathbb{R} : x > 0\}$:

Note that $0 \notin X$, and so $\delta(x, y)$ is well-defined for all $x, y \in X$.

$$(M1) \text{ If } x = y \text{ we have } \delta(x, y) = \left| \frac{1}{x} - \frac{1}{y} \right| = 0,$$

conversely if $\delta(x, y) = 0$, then $\left| \frac{1}{x} - \frac{1}{y} \right| = 0$, and so $\frac{1}{x} - \frac{1}{y} = 0$, that is $\frac{1}{x} = \frac{1}{y}$, and thus $x = y$.

$$(M2) \delta(x, y) = \left| \frac{1}{x} - \frac{1}{y} \right| = \left| \frac{1}{y} - \frac{1}{x} \right| = \delta(y, x).$$

$$(M3) \delta(x, z) = \left| \frac{1}{x} - \frac{1}{z} \right| = \left| \frac{1}{x} - \frac{1}{y} + \frac{1}{y} - \frac{1}{z} \right| \leq \left| \frac{1}{x} - \frac{1}{y} \right| + \left| \frac{1}{y} - \frac{1}{z} \right| = \delta(x, y) + \delta(y, z)$$

$$(b) B[4; 1] = \{x \in X : \delta(4, x) \leq 1\} = \{x \in X : \left| \frac{1}{4} - \frac{1}{x} \right| \leq 1\} = \{x \in X : -1 \leq \frac{1}{4} - \frac{1}{x} \leq 1\} = \{x \in X : \frac{1}{x} \leq \frac{5}{4}\} = [\frac{4}{5}, \infty).$$

$$(c) x_n \rightarrow a \text{ w.r.t. } \delta \text{ iff } \delta(x_n, a) \rightarrow 0 \text{ w.r.t. } d_E \text{ iff } \left| \frac{1}{x_n} - \frac{1}{a} \right| \rightarrow 0 \text{ w.r.t. } d_E \text{ iff } \frac{1}{x_n} - \frac{1}{a} \rightarrow 0 \text{ w.r.t. } d_E \text{ iff } x_n \rightarrow a \text{ w.r.t. } d_E. \quad [6+4+3=13]$$

3. ($\implies$) If $\text{bd}(A) = \emptyset$, then $A \cap \text{bd}(A) = A \cap \emptyset = \emptyset$, so A is open,
and again if $\text{bd}(A) = \emptyset$, then $\overline{A} = A \cup \text{bd}(A) = A \cup \emptyset = A$, so A is closed.
($\impliedby$) If A is closed, $\overline{A} = A$ and also if A is open, then A' is closed and so $\overline{A'} = A'$,
thus $\text{bd}(A) = \overline{A} \cap \overline{A'} = A \cap A' = \emptyset$. [6]
4. (a) We are assuming $x \in X$ and so $x \in \overline{A}$. That there exists a sequence (a_n) such
that $a_n \in A$ for all n and $a_n \rightarrow x$ is the definition of $x \in \overline{A}$.
(b) Such an N exists because $r > 0$ and $a_n \rightarrow x$.
(c) $a_N \in B(x; r) \cap A$ because $d(a_N, x) < r$ and $a_N \in A$.
(d) By assumption, $A \cap B(z; r) \neq \emptyset$ for any $z \in X$ and $r > 0$. This applies using
 $z = x$ and $r = \frac{1}{n}$.
(e) From line (6), we have a sequence (c_n) such that $c_n \in A$ for all n and $c_n \rightarrow x$,
because $d(c_n, x) < \frac{1}{n} \rightarrow 0$.
(f) In line (5), we began with $x \in X$ and in line (7) showed $x \in \overline{A}$, so this shows
 $X \subseteq \overline{A}$ and since the $\overline{A} \subseteq X$ always holds, this shows equality $\overline{A} = X$.
(g) Yes: take $(-\infty, 0) \cup (0, \infty)$, for example. [7 $\times$ 2=14]
5. We first check that for a, b, c, d irrational, the set
 $A = \{(x, y) \in \mathbb{Q} \times \mathbb{Q} \mid a < x < b \text{ and } c < y < d\}$ is clopen in $\mathbb{Q} \times \mathbb{Q}$:
 A is closed: Take sequence $(x_n, y_n) \rightarrow (x, y)$ in $\mathbb{Q} \times \mathbb{Q}$ and $(x_n, y_n) \in A$ for all n ,
then $a \leq x_n \leq b$ for all n , and so $a \leq x \leq b$ but since a and b are irrational, while
 x is rational, we have $a < x < b$. Similarly, $c < y < d$.
 A is open: Note $A = (\mathbb{Q} \times \mathbb{Q}) \cap \{(x, y) \in \mathbb{R} \times \mathbb{R} \mid a < x < b \text{ and } c < y < d\}$ and the
second of these sets is open in $\mathbb{R} \times \mathbb{R}$.
Further, each open set U in $\mathbb{Q} \times \mathbb{Q}$ is union of these: for each $(x, y) \in U$ there exists
irrational $r > 0$ such that $B((x, y), r) \subseteq U$. Then a square of side $\frac{r}{2}$ centred at (x, y)
is contained in U . [4]